

Scope of Work (SOW): Field Logistics & Travel Management System

1. System Overview & Authentication

A centralized web-based platform to manage field logistics, travel requests, and accommodation for ground staff, streamlining operations and fulfillment for the admin team.

- **Authentication & RBAC:** Secure login system with Role-Based Access Control distinguishing between the Ground Team (approx. 100 users) and the Admin Team (approx. 10 users).
- **Theme Preferences:** Native Light/Dark mode toggle for user comfort across all devices.

2. Campaign & Project Management

- **Project Lifecycle Control:** Admins possess full CRUD (Create, Read, Update, Delete) capabilities to add, edit, delete, or hide specific projects and campaigns based on operational status.
- **Request Tagging:** Ground staff are required to select an active project/campaign from a dropdown when submitting any travel or accommodation request, ensuring all logistics are tied to a specific operational goal.
- **Campaign-Level Financials:** Enables admins to track total campaign details, yielding reports on the total number of personnel involved in a specific project, alongside aggregate travel, hotel, and local transit costs.

3. Request Management Module (Ground Team)

- **Multi-Category Request Form:** Ground staff can initiate requests for long-distance travel (flight/train/bus), local cabs, and hotel accommodations.
- **Group Requests & Co-Travelers:** A single employee can submit a travel or accommodation request and "tag" other employees joining them (e.g., booking one cab for 4 specific people, or a train journey for a group).
- **Automated Conflict & Double-Booking Detection:**
 - When a request is initiated, the system validates the dates against the existing itineraries of the requester and all tagged co-travelers.
 - If any included employee already has an overlapping active request or approved booking in the system, it flags a conflict error to prevent double-booking.
- **Intelligent Accommodation Matching (Co-Stay):**
 - When a hotel is requested, the system queries active bookings in that city.
 - If another employee of the same gender is already booked during those dates, the system prompts the requester with the option to share the room at the same hotel, or explicitly request a different room or a different hotel.
 - Cross-gender requests automatically enforce the separate room policy.

4. Fulfillment & Booking Module (Admin Team)

- **Action Queue & Status Tracking:** A dashboard to view all incoming requests categorized by status (Pending, Partially Approved, Booked, Rejected).
- **Partial & Selective Approvals:** When an admin reviews a group request (e.g., 4 people requested for a cab/flight), the admin can approve the entire group OR selectively approve/reject specific individuals within that request based on operational needs.
- **AI-Powered Automated Ticket Extraction:**
 - Admins bypass manual data entry by uploading the raw booking PDF, screenshot, or voucher.
 - The system utilizes an integrated LLM to automatically parse and extract key data (PNR, booking dates, travel times, hotel names, passenger names) and populate the database fields instantly.
- **Automated Notifications:** The system updates the status to Booked upon document upload and automatically notifies the approved ground staff members (via email integration) with their respective tickets and details.

5. Employee Management & Hierarchical Profiles

- **Directory Management:** Admins can Add, Edit, or Delete employee profiles, managing core details like Name, Gender, Contact, and ID Proofs.
- **Ground Staff Role Definitions:** Employees are assigned specific operational designations—**Executives**, **Team Leads (TL)**, and **Managers**—allowing for clear hierarchical organization and future routing of approval workflows.
- **Comprehensive Travel History:** Each employee profile features a timeline view showing a chronological history of their movements (minimum 2+ months lookback). This tracks all travel dates, modes of transport, cab companions, and hotel stay locations.

6. Admin Dashboard & Reporting

- **Live Operations Overview:** A high-level visual dashboard displaying:
 - Count of active ground staff currently deployed, grouped by state/location and active campaign.
 - Queue of pending requests requiring immediate action.
 - Recent booking activity and system alerts.
- **Cost Analytics:** Aggregated views of operational expenditures filtered by project, location, or specific timeframes.

7. System Security & Comprehensive Audit Logging

- **Universal Activity Ledger:** The system maintains an immutable, background audit trail of every action taken by any user and all Admins.
- **Data Tracking:** It records exactly who made a request, when it was made, who approved/rejected it (including partial rejections), and when documents were uploaded or profiles were edited. This ensures total accountability and a clear paper trail for all logistical movements and administrative decisions.